

1. Executive Summary

RailVue AI is an end-to-end operational train dispatch and passenger intelligence platform engineered specifically for the complexities of the **Indian Railways** network. It transitions train operations from static timetable tracking to proactive, machine-learning-driven **Estimated Time of Arrival (ETA)** predictions that continuously account for real-time section speeds, junction bottlenecks, preceding train density, and adverse meteorological conditions.

10.37 min
TUNED XGBOOST MAE

42.3%
ERROR REDUCTION VS
TIMETABLE

84.1%
DELAY RISK CLASSIFIER ACC.

1,500+
INDEXED RAIL DIRECTORY

Key Operational Benchmark: The standard static timetable schedule incurs an average arrival error of **17.96 minutes**. RailVue AI's tuned gradient-boosted engine slashes prediction error down to **10.37 minutes**, delivering actionable punctuality intelligence to railway section controllers and daily passengers.

2. Architectural Breakthrough: The Delay-Deviation Target

In long-distance railway systems, training models to predict absolute remaining duration directly leads to *metric inflation*. Because journey distances range from 50 km to over 1,500 km, trip length variance dominates the Total Sum of Squares (SS_{tot}), yielding artificially inflated R^2 values (>0.95) that conceal poor real-world delay tracking.

The Delay Deviation Formulation (Δ_{delay})

RailVue AI solves this by reframing the core prediction objective to **Delay Deviation from the Official Timetable**:

```
[Target Definition]
 $\Delta_{delay}$  = Actual Arrival Delay - Current Observed Delay
model_target_column = "delay_deviation_minutes"

[Serving-Time ETA Reconstruction]
Dynamic ETA = Scheduled Arrival Time + Current Delay +  $\Delta_{predicted}$ 
```

By isolating dynamic delay drift, the models focus strictly on stochastic operational factors: track congestion, signal halts, dispatch priorities, and weather restrictions.

3. Anti-Leakage Data Pipeline & Feature Vector

To guarantee true generalization on unseen journeys, the dataset pipeline enforces three engineering guardrails:

- **Journey-Aware Group Splitting:** Dataset rows are grouped using GroupKFold based on unique journey_id. Multiple snapshots of the same train run are strictly isolated to either the training or test set, preventing temporal leakage.
- **Train-Fold Only Historical Priors:** Corridor congestion scores and station dwell variance are calculated strictly on training folds and mapped to test folds.
- **Physics-Informed Sanity Bounds:** Serving-time ETAs are bounded by physical corridor speed limits (160 km/h for Vande Bharat, 130 km/h for Rajdhani/Shatabdi), preventing negative transit times or impossible velocities.

Feature Vector (18 Engineered Signals)

Spatial & Telemetry Features:

- distance_remaining_km
- journey_progress_pct
- current_speed_kmph
- current_delay_minutes
- distance_to_next_station_km

Network & Environmental Signals:

- scheduled_remaining_time_minutes
- corridor_congestion_score
- weather_score & rainfall_mm
- signal_delay_score
- historical_avg_delay_minutes

4. Empirical Performance & Model Evaluation

Evaluated on a strictly held-out test split of **515 unseen journey snapshots** across 37 distinct train runs:

Model / Pipeline	MAE (Min)	RMSE (Min)	R ² (Absolute)	R ² (Delay Signal)	Error Reduction
Timetable Schedule Baseline	17.96	23.41	0.9934	0.0000	BASELINE
Random Forest Regressor	10.62	13.90	0.9977	0.4810	+40.9%
Tuned XGBoost Regressor (Production)	10.37	13.65	0.9977	0.4993	+42.3%

Performance by Journey Distance Segment

Segment Haul	Distance Range	Sample Count	XGBoost MAE	Schedule Baseline MAE	Accuracy Gain
Short-Haul	< 100 km	172	6.44 min	12.80 min	+49.7%
Medium-Haul	100 – 300 km	171	11.23 min	18.95 min	+40.7%
Long-Haul	> 300 km	172	14.88 min	22.10 min	+32.7%

Understanding the Two R² Scores: On absolute travel time, R² is **0.9977** because trip duration variance across 1,500 km dominates the denominator ($SS_{\text{tot}} = 42,352,134$). On the isolated delay signal (Δ_{delay}), R² is **0.4993** (explaining ~50% of real-world stochastic delay variations on unseen train journeys, compared to 0% for the timetable).

5. Operational Delay-Risk Classifier

Alongside regression, RailVue AI equips controllers with a 3-class Random Forest delay-risk classifier:

Risk Thresholds:

- **ON_TIME** : Delay deviation ≤ 10 minutes
- **MINOR_DELAY** : Delay deviation 10 – 30 minutes
- **MAJOR_DELAY** : Delay deviation > 30 minutes

Validation Results:

- **Accuracy:** 84.1%
- **Macro F1-Score:** 0.772
- **False-Alarm Guardrail:** Only 1 misclassification out of 113 on-time journeys flagged as major delay.

6. Human-in-the-Loop Explainability Engine

Predictions are never black-box numbers. The explainability layer (`backend/ml/explainability.py`) attributes ETA variance to physical operational drivers:

- **Track Congestion:** Preceding train density in critical block sections (e.g., Kanpur Central – Prayagraj – Pt. DD Upadhyaya).
- **Signal Dwell:** Waiting time for line and platform clearances at busy junction loops.
- **Weather Restrictions:** Mandatory speed reductions for fog or heavy rainfall.
- **Speed Recovery Bonus:** Timetable slack time allowing high-speed trains to recover lost minutes.

7. Full-Stack Application Architecture & UI Innovations

- 1 **Data & Telemetry Engine:** Indian Railways station directory (1,500+ trains) with real-time GPS telemetry simulation.
- 2 **FastAPI Intelligence Core (Port 8000):** 15-second simulation ticker serving dual regression and classification inference.
- 3 **Dynamic Route Geometry SVG Map:** 50+ real Indian Railways station coordinates with dynamic auto-fit bounding box and train marker snapping.
- 4 **Continuous Timeline Track:** Centered vertical track connector with station node status (completed gold / upcoming slate).
- 5 **Corridor Routing Engine (Find Trains):** Multi-corridor trunk matching (Grand Chord, Western, Konkan, Central) with automatic station code resolution.
- 6 **Production Cloud Resilience:** Vercel SPA routing with `vercel.json`, React `ErrorBoundary`, and automatic offline simulation fallback.

8. Project Resources & Live Deployment Verification

Resource	URL / Location	Status
Live Vercel Production Web App	https://rail-collector-zeta.vercel.app	ACTIVE (HTTP 200)
GitHub Repository (Origin)	https://github.com/SumitKumar-990/rail_collector	SYNCED (MAIN BRANCH)
Trained Production Model	<code>backend/models/eta_xgboost.json</code>	VERSIONED ARTIFACT
Delay Risk Classifier	<code>backend/models/delay_risk_classifier.pkl</code>	VERSIONED ARTIFACT
Local FastAPI Backend API	http://127.0.0.1:8000/docs	SWAGGER OPENAPI